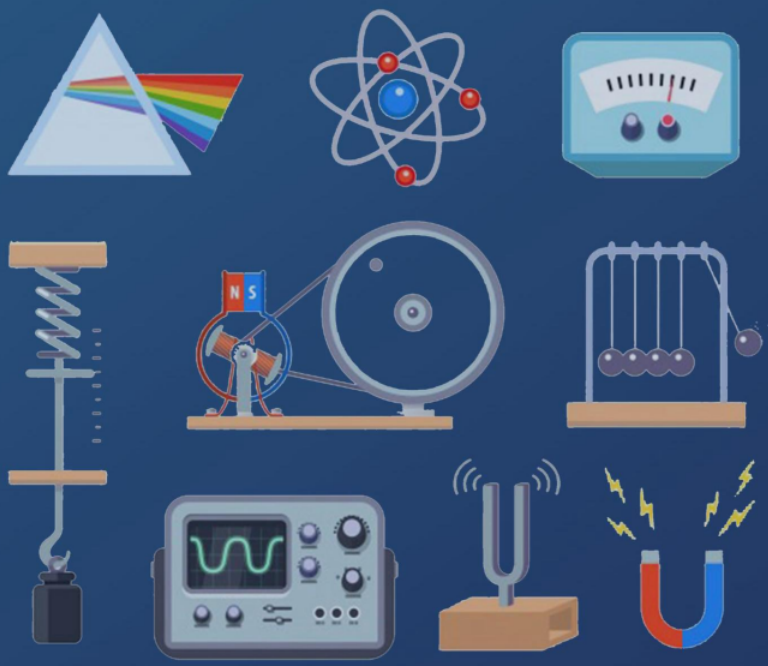
PHYSICS

Ray Optics



SHEET

eSaral.com

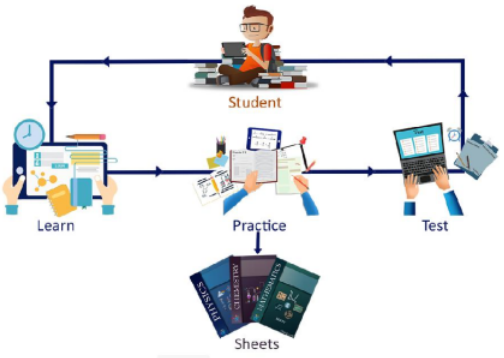
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

S.No	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction and Reflection of Light				
2	Object and Image				
3	Plane mirror	1-3		1-3	
4	Plane Mirrors at an Angle	4	1, 12	4	
5	Introduction to Spherical Mirrors				
6	Paraxial approximation and Focal length				
7	Mirror Equation	5-8			
8	Mirror Equation- Practice	9-12, 13	13		1, 26, 30
9	v-u graphs of Mirrors			5	
10	Newton's Formula	14	2, 9-11, 14		
11	Longitudinal Magnification and Image Velocity	15, 16			2
12	Refraction		3		3
13	Refraction - Practice			6	
14	Apparent Depth	17, 18			4
15	Refraction through Parallel slab			7-8	
16	TIR	19-22		9-10	5, 27, 31
17	TIR Applications and Natural Phenomena	23, 24			
18	Angle of Deviation	25			
19	Refraction through a prism	26			25
20	Minimum and Maximum Angle of Deviation	27-30	19	11-14	6, 14-16
21	Thin prism				
22	Refraction from Curved Surface	31, 32	20	15	7
23	Refraction by lens		15		8
24	Lens Maker's Formula	33, 34	4, 17	16	28, 32
25	Applying Lens Formula	35	5-7	17	
26	Thin Lens - Longitudinal Magnification and Velocity of Image	36	21		29
27	Power of Lens	37-40			33
28	Silvering of Lens			18	
29	Dispersion	41-44		19-20	9-10
30	Human Eye, Defects and its Corrections	45-49		21-22	17-21
31	Simple Microscope	50			34
32	Compound Microscope				
33	Telescope			23	22-24
34	After full Chapter	51	8, 16, 18	24-31	11-13, 35

Plane mirror

Q1. A tall man of height 6 feet, want to see his full image. Then required minimum length of the mirror will be :-

(A) 12 feet (B) 3 feet

(C) 6 feet (D) Any length

Q2. A man runs towards stationary plane mirror at a speed of 15 m/s. What is the speed of his image with respect to mirror :-

(A) 7.5 m/s (B) 15 m/s

(C) 30 m/s (D) 45 m/s

Q3. When a ray of light is incident normally on a plane mirror then the angle of reflection will be :

(A) 0° (B) 90° (C) 180° (D) – 90°

Plane Mirrors at an Angle

Q4. Two plane mirrors are lying perpendicular to each other, there is lamp in between mirrors. Then No. of images of lamp will be :

(A) 3 (B) 4 (C) 5 (D) 6

Mirror Equation

Q5. An object of height 7.5 cm is placed in front of a convex mirror of radius of curvature 25 cm at a distance of 40 cm. The height of the image should be -

(A) 2.3 cm (B) 1.78 cm

(C) 1 cm (D) 0.8 cm

Q6. An object of height 1.5 cm is situated at a distance of 15 cm from a concave mirror. The concave mirror forms its real image of height 3.0 cm. The focal length of concave mirror is-

(A) – 10 cm (B) – 20 cm

(C) 20 cm (D) 30 cm

Q7. If an object is placed 10 cm in front of a concave mirror of focal length 20 cm, the image will be:

(A) Diminished, upright, virtual

(B) Enlarged, upright, virtual

(C) Diminished, inverted, real

(D) Enlarged, upright, real

Q8. A convex mirror has a radius of curvature of 22 cm. If an object is placed 14 cm away from the mirror then its image is formed at -

(A) 6.2 cm on the front side of the mirror.

(B) 6.2 cm on the back side of the mirror.

(C) 51.3 cm on the front of the mirror.

(D) 51.3 cm on the back side of the mirror.

Mirror Equation- Practice

Q9. The focal length of a concave mirror is 12 cm. Where should an object of length 4 cm be placed, so that a real image of 1 cm length is formed ?

(A) 48 cm (B) 3 cm

(C) 60 cm (D) 15 cm

Q10. An object is lying at a distance of 90 cm from a concave mirror of focal length 30 cm. The position and nature of image formed by it will be :

(A) 45 cm, of the size of object

(B) 90 cm, smaller than object

(C) 30 cm, bigger than object

(D) – 45 cm smaller than object

Q11. An object is situated at a distance of 15 cm from a convex mirror of focal length 30 cm. The position from mirror of the image formed by it will be-

(A) – 5 cm (B) 10 cm

(C) – 10 cm (D) + 5 cm

Q12. A virtual image three times the size of the object is obtained with a concave mirror of radius of curvature 36 cm. The distance of the object from the mirror is -

(A) 5 cm (B) 12 cm
(C) 10 cm (D) 20 cm

Q13. The focal length of a concave mirror is 50 cm. Where an object be placed so that its image is two times magnified, real and inverted :-

(A) 75 cm (B) 72 cm
(C) 63 cm (D) 50 cm

Newton's Formula

Q14. A diminished virtual image can be formed only in

(A) Plane mirror
(B) A concave mirror
(C) A convex mirror
(D) Concave- parabolic mirror

Longitudinal Magnification and Image Velocity

Q15. A square wire of side 3 cm is placed at a distance of 25 cm from a concave mirror of focal length 10 cm. The centre of the square is at the axis of the mirror and the plane is normal to the axis. The area enclosed by the image of the wire is -

(A) 4 cm^2 (B) 6 cm^2
(C) 16 cm^2 (D) 36 cm^2

Q16. A point object is moving on the principal axis of a concave mirror of focal length 24 cm towards the mirror. When it is at a distance of 60 cm from the mirror, its velocity is 9 cm/sec. What is the velocity of the image at that instant -

(A) 5 cm/sec. (B) 12 cm/sec
(C) 4 cm/sec (D) 9 cm/sec

Apparent Depth

Q17. Velocity of light in glass, whose refractive index w.r.t. air is 1.5, is 2×10^8 m/s. In a certain liquid the velocity of light is found to be 2.5×10^8 m/s. The refractive index of liquid w.r.t. air is :-

(A) 0.64 (B) 0.80
(C) 1.20 (D) 1.44

Q18. A vessel is half filled with a liquid of refractive index μ. The other half of the vessel is filled with an immiscible liquid of refractive index 1.5μ. The apparent depth of vessel is 50% of the actual depth. The value of μ is :-

(A) 1.6 (B) 1.67
(C) 1.5 (D) 1.4

Total Internal Reflection

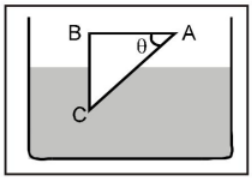
Q19. A point source of light is place 4 m below the surface of water of refractive index 5/3. The minimum diameter of a disc which should be placed over the source on the surface of water to cut-off all light coming out of water is ($\mu = 5/3$)

(A) 2 m (B) 6 m
(C) 4 m (D) 3 m

Q20. The wavelength of light in two liquids 'x' and 'y' is 3500 Å and 7000 Å, then the critical angle for light travelling from x to y will be.

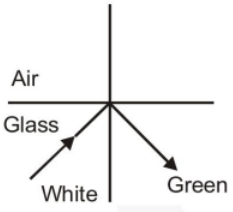
(A) 60° (B) 45°
(C) 30° (D) 15°

Q21. A glass prism ($\mu = 1.5$) is dipped in water ($\mu = 4/3$) as shown in figure. A light ray is incident normally on the surface AB. It reaches the surface BC after totally reflected, if



(A) $\sin\theta \geq 8/9$
(B) $2/3 < \sin\theta < 8/9$
(C) $\sin\theta \leq 2/3$
(D) It is not possible

Q22. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains :



(A) Yellow, orange, red
(B) Violet, indigo, blue
(C) All colours
(D) All colours except green

TIR Applications and Natural Phenomena

Q23. Which of the following is used in optical fibres
(A) Total internal reflection
(B) Scattering
(C) Diffraction
(D) Refraction

Q24. Brilliance of diamond is due to :-
(A) Scattering
(B) Diffraction
(C) Reflection
(D) Total internal reflection

Angle of Deviation

Q25. A ray is incident at 30° angle on plane mirror. What will be deviation after reflection from mirror.
(A) 120° (B) 60°
(C) 30° (D) 45°

Refraction through a prism

Q26. A ray of light is incident at 60° on a prism of refracting angle 30°. The emerging ray is at an angle 30° with the incident ray. The value of refractive index of the prism is :-

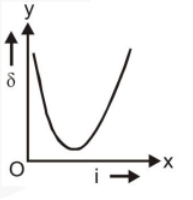
(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$

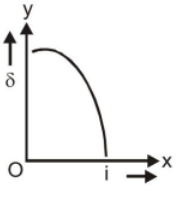
(C) $\sqrt{3}$ (D) $\frac{2}{\sqrt{3}}$

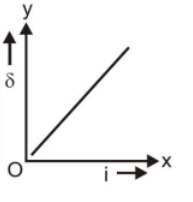
Minimum and Maximum Angle of Deviation

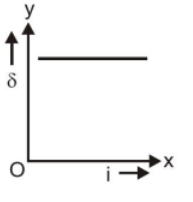
Q27. The angle of the prism is 60° and minimum deviation of 30°, then the angle of incidence for minimum deviation is :-
(A) 30° (B) 45°
(C) 25° (D) 60°

Q28. For a prism, graph between angle of deviation (δ) and angle of incidence will be:-

(A) 

(B) 

(C) 

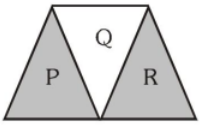
(D) 

Q29. The angle of minimum deviation measured with a prism is 30° and the angle of prism is 60°. The refractive index of prism material is :-

(A) $\sqrt{2}$ (B) 2

(C) $\frac{3}{2}$ (D) $\frac{4}{3}$

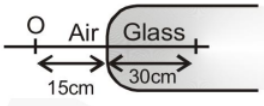
Q30. A given ray of light suffers minimum deviation in an equilateral prism P. Additional prism Q and R of identical shape and of the same material as P are now added as shown in the figure. The ray will suffer:-



(A) Greater deviation
(B) No deviation
(C) Same deviation as before
(D) Total internal reflection

Refraction from Curved Surface

Q31. A point object O is placed in front of a glass rod having spherical end of radius of curvature 30 cm. The image formed would be ..... from spherical end.



(A) 30 cm left
(B) infinity
(C) 1 cm to the right
(D) 18 cm to the left

Q32. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at a point O and PO = OQ. The distance PO is equal to :-

(A) R (B) 3 R (C) 2 R (D) 5 R

Lens Maker's Formula

Q33. Focal length of convex lens is f. If this lens is cut along parallel to its length in two equal parts then focal length of its half part will be :

(A) f/2 (B) f (C) $\frac{3f}{2}$ (D)2f

Q34. The focal length of a convex lens is 10 cm and its refractive index is 1.5. If the radius of curvature of one surface is 7.5 cm, the radius of curvature of the second surface will be :

(A) 7.5 cm (B) 15.0 cm
(C) 75 cm (D) 5.0 cm

Applying Lens Formula

Q35. An object is put at a distance of 15 cm from a convex lens of focal length 10cm.If a real image is formed. Its distance from the lens will be :

(A) 15 cm (B) 20 cm
(C) 25 cm (D) 30 cm

Thin Lens - Longitudinal Magnification and Velocity of Image

Q36. A point object is placed at a distance of 10 cm and its real image is formed at a distance of 20 cm from a convex lens. If the object is moved by 0.1 cm towards the mirror, the image will shift by about.

(A) 0.4 cm away from the lens
(B) 0.4 cm towards the lens
(C) 0.8 cm away from the lens
(D) 0.8 cm towards the lens

Power of Lens

Q37. A convex lens is in contact with concave lens. The magnitude of the ratio of their focal length of convex & concave lens is 2/3. Their equivalent focal length is 30cm. What are their individual focal lengths ?

(A) 50,–75 (B) –10, 15
(C) –50, 75 (D) 10,–15

Q38. Two convex lens of focal length 20 cm and 25 cm are placed in contact with each other, then power of this combination is :-

(A) + 1 D (B) + 9 D
(C) – 1 D (D) – 9 D

Q39. Lenses of powers 3D and -5D are combined to from a compound lens. An object is placed at a distance of 50 cm from this lens. Calculate the position of its image.

(A) -10cm (B) +10cm

(C) -25cm (D) + 25cm

Q40. An equiconvex lens has a power of 5 dioptre. If it is made of glass of refractive index 1.5. then radius of curvature of its each surface will be ?

(A) 20cm (B) 10cm

(C) 40cm (D) ∞

Dispersion

Q41. Rainbow is formed due to :-

(A) Scattering & refraction

(B) Total internal reflection & dispersion

(C) Reflection only

(D) Diffraction and dispersion

Q42. A plane glass slab is kept over various coloured letters; the letter which appears least raised is

(A) blue (B) voilet

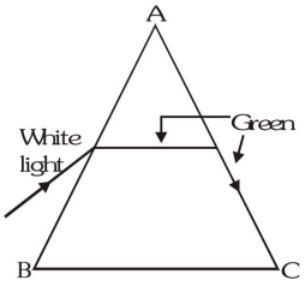
(C) green (D) red

Q43. Prism angle of glass prism is 10°. It's refractive index of red and violet colour is 1.51 and 1.52 respectively. Then its dispersive power will be :

(A) 0.015 (B) 0.020

(C) 0.011 (D) 0.019

Q44. White light is incident on face AB of a glass prism.The path of the green component is shown in the figure. If the green light is just totally internally reflected at face AC as shown, the light emerging from face AC will contain :



(A) yellow, orange and red colours

(B) violet, indigo and blue colours

(C) all colours

(D) all colours except green

Human Eye, Defects and its Corrections

Q45. A person can not see the objects beyond 50 cm. The power of a lens to correct this vision will be-

(A) +2D (B) –2D

(C) +5D (D) 0.5 D

Q46. A person can not see clearly an object kept at a distance beyond of 100 cm. Find the nature and the power of lens to be used for seeing clearly the object at infinity.

(A) 1D (B) – 1D

(C) 2D (D) – 2D

Q47. A person can see clearly only upto a distance of 25 cm. He wants to read a book placed at a distance of 50 cm. What kind of lens does he require for his spectacles and what must be its power?

(A) Concave, – 1.0 D

(B) Convex, + 1.5 D

(C) Concave, – 2.0 D

(D) Convex, + 2.0 D

Q48. A long sighted person has a minimum distance of distinct vision of 50 cm. He wants to reduce it to 25 cm. He should use a:

(A) Concave lens of focal length 50 cm

(B) Convex lens of focal length 25 cm

(C) Convex lens of focal length 50 cm

(D) Concave lens of focal length 25 cm

Q49. To remove myopia (short sightedness) a lens of power 0.66D is required. The distant point of the eye is approximately :

(A) 100 cm (B) 151.5 cm

(C) 50 cm (D) 25 cm

Simple Microscope

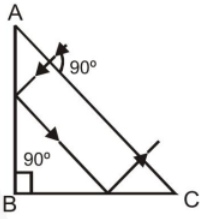
Q50. The focal length of convex lens is 2.5 cm. Its magnifying power for minimum distance of distinctvision will be :-

(A) 25 (B) 52

(C) 11 (D) 1.1

Miscellaneous

Q51. A ray falls on a prism ABC (AB = BC) and travels as shown in figure. The minimum refractive index of the prism material should be :



(A) $\frac{4}{3}$ (B) $\sqrt{2}$

(C) 1.5 (D) $\sqrt{3}$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\theta = 120°$ then total number of images formed by the mirror system of object not may be :-

(A) 2 (B) 3 (C) 4 (D) 5

Q2. A concave mirror forms an image of the sun at a distance of 12 cm from it.

(A) The radius of curvature of this mirror is 6 cm.

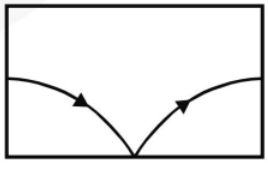
(B) To use it as a shaving mirror, it can be held at a distance of 8-10 cm from the face

(C) If an object is kept at a distance of 24 cm from it, the image formed will be of the same size as the object

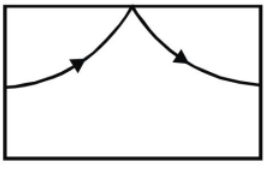
(D) All the above alternatives are correct.

Q3. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following figures may represent the path of a ray of light inside the liquid?

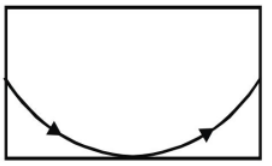
(A)



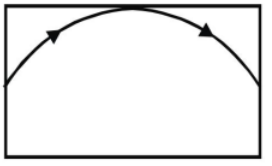
(B)



(C)



(D)



Q4. A lens is formed by a material having refecting index $\frac{3}{2}$ & radii of curvature 20cm & 10 cm. Then choose the correct option(s):-

(A) If lens is concavo-convex & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm.

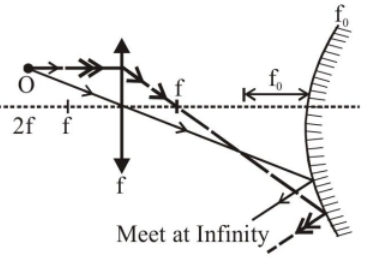
(B) If lens is concavo-convex & light is falling on the surface having radius of curvature 10 cm, then focal length is – 40 cm.

(C) If lens is convexo-concave & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm.

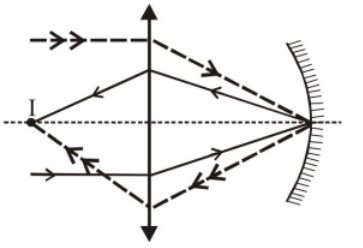
(D) If lens is convexo-concave & light is falling on the surface having radius of curvature 10 cm, then focal length is – 40 cm.

Q5. Choose the incorrect ray diagram(s). All the rays shown are paraxial. [↕ denotes converging lens and | denotes diverging lens] :

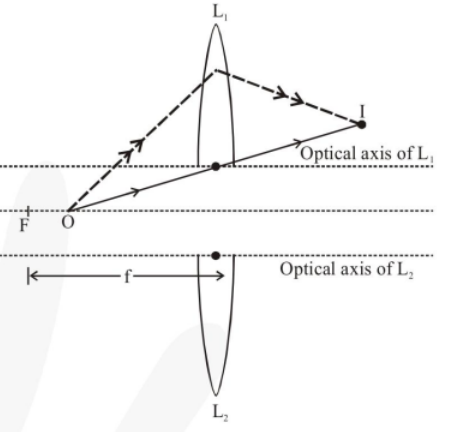
(A)



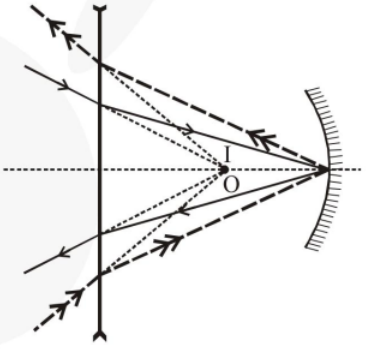
(B)



(C)

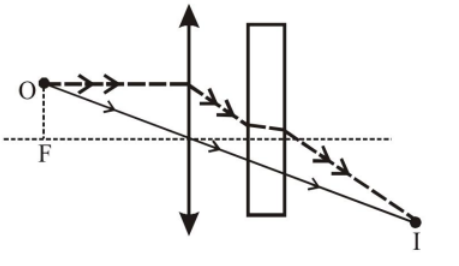


(D)

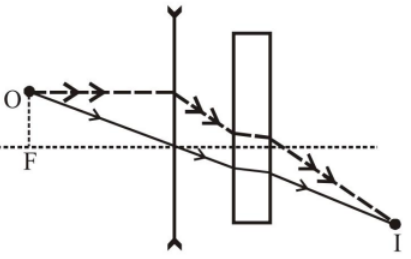


Q6. Choose incorrect ray diagram [↕ denotes converging lens and | denotes diverging lens] All symbols have their usual meaning and all the rays shown are paraxial. (focal length of each lens is F)

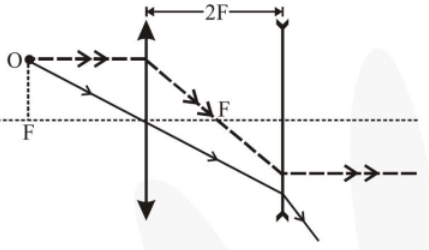
(A)



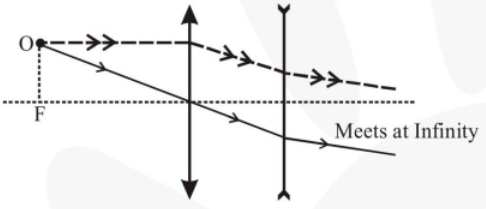
(B)



(C)



(D)



Q7. A man wanted to get a picture of a Zebra. He photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.

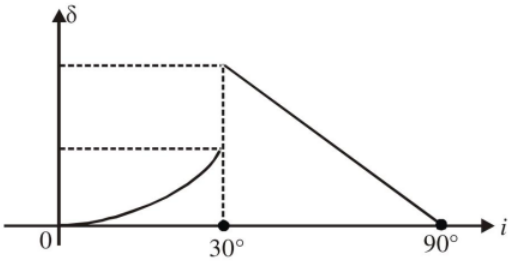
(A) the image will look like a white donkey on the photograph.

(B) the image will look like a Zebra on the photograph.

(C) the image will be more intense compared to the case in which no such glass is used.

(D) the image will be less intense compared to the case in which no such glass is used.

Q8. Figure shows graph of angle of deviation v/s angle of incidence for a light ray. Incident ray goes from medium 1 (μ_1) to medium 2 (μ_2). Mark the **correct** option(s).



(A) $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

(B) Critical angle is 30°

(C) $\mu_1 > \mu_2$

(D) Maximum deviation is 120°

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 9 to 11

A real object is placed perpendicular to the principal axis of a concave mirror at a position (1) such that the image formed is real with a magnification 2. The object is now shifted to another position (2) at a distance 15 cm from the position (1) and a real image is obtained with a magnification 8. Shifting the position of object from (2) to a third position (3) gives an image with magnification 1/2.

Q9. Position of the image when the object is at position (3) is :-

(A) between centre of curvature and ∞

(B) between focus and centre of curvature

(C) at focus

(D) between pole and focus

Q10. Focal length of the mirror is :-

(A) – 60 cm

(B) –40 cm

(C) –30 cm

(D) –25 cm

Q11. Distance of position (1) of the object from pole and that of position (2) of object from pole are , respectively :-

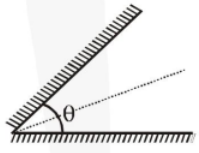
(A) 60 cm, 75 cm (B) 75 cm, 60 cm

(C) 60 cm, 45 cm (D) 45 cm, 30 cm

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12. Angle between two mirrors 'θ' and location of object is given in column I and some possible number of images are given in column II.



Column –I	**Column –II**
(A) θ = 30° and object is on bisector	(P) 9
(B) θ = 36° and object is on bisector	(Q) 11
(C) θ = 22.5° and object is on bisector	(R) 5
(D) θ = 72° and object is not on bisector	(S) 15

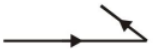
Q13.

Column-I	**Column-II**
(A) An object is placed at a distance equal to focal length from pole before convex mirror	(P) Maginification is (∞)
(B) An object is placed at focus before a concave mirror	(Q) Magnification is (0.5)
(C) An object is placed at the centre of curvature before a concave mirror.	(R) Magnification is (1/3)
(D) An object is placed at a distance equal to radius of curvature before a convex mirror.	(S) Magnification is (–1)

Q14. An object O is kept perpendicular to the principal axis of a spherical mirror. In column I, coordinate of, object, u with sign in cm, type of mirror and its focal length in cm is given

Column I	**Column II**
(A) u = –18, concave mirror, 12	(P) Image is real
(B) u = –12, concave mirror, 18	(Q) Image is virtual
(C) u = –8, convex mirror, 10	(R) Image diminished
(D) u = –10, convex mirror, 8	(S) Image is enlarged
	(T) Image is inverted

Q15. Four rays of light parallel and above to optic axis and their path after passing through an optical system are shown in column-I. Match the corresponding optical instrument from column-II :

Column-I	Column-II
(A) 	(P) Convex lens
(B) 	(Q) Concave lens
(C) 	(R) Convex mirror
(D) 	(S) Concave mirror

Q16. Column–II shows the optical phenomenon that can be associated with optical components kept in air given in column–I. Note that column–I may have more than one matching options in column–II.

Column–I	Column–II
(A) Convex mirror	(P) Dispersion
(B) Converging lens	(Q) Deviation
(C) Thin prism	(R) Real image of real object
(D) Glass slab	(S) Virtual images of real object

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. What is the refractive index of material of a plano-convex lens, if the radius of curvature of the convex surface is 10 cm and focal length of the lens is 30 cm ?

Q18. Prism angle of a prism is 10°. Their refractive index for red and violet colour is 1.51 and 1.52 respec tively. Then find the dispersive power.

SUBJECTIVE TYPE

Q19. The angle of minimum deviation for prism of angle $\frac{\pi}{3}$ is $\frac{\pi}{6}$. Calculate the velocity in the material of the prism, if the velocity of light in vacuum is $3 \times 10^8\ ms^{-1}$.

Q20. A beam of light converges towards a point O, behind a convex mirror of focal length 20 cm. Find the nature and position of image if the point O is :-

(a) 10 cm behind the mirror

(b) 30 cm behind the mirror.

Q21. A magnifying lens has a focal length of 10 cm.

(a) Where should the object be placed if the image is to be 30 cm from the lens ?

(b) What will be the magnification ?

SINGLE CORRECT TYPE

Q1. A man on ground of height 170 cm wants to see his complete image in a plane mirror (while standing) . His eyes are at a height of 160 cm from the ground. choose incorrect statement.

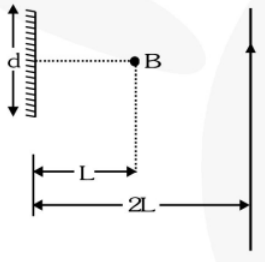
(A) Minimum length of the mirror = 80 cm

(B) Minimum length of the mirror = 85 cm

(C) Bottom of the minimum mirror should be at a height 85 cm from ground.

(D) Top of the minimum mirror should be 5 cm above eye level.

Q2. A point source of light B, placed at a distance L in front of the centre of plane mirror of width d, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 2L from it as shown. The greatest distance over which he can see the image of the light source in the mirror is :



(A) d/2 (B) d

(C) 2d (D) 3d

Q3. A point object approaches a plane mirror with a speed of 10 ms^{-1}, while the image recedes away from the mirror with a speed of 6 ms^{-1} (as seen by stationary observer). The direction and magnitude of the velocity of mirror is :-

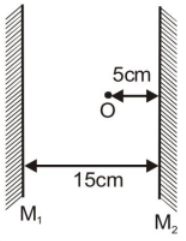
(A) towards the object, 8 ms^{-1}

(B) towards the image, 6 ms^{-1}

(C) away from the object, 8 ms^{-1}

(D) away from the object, 2 ms^{-1}

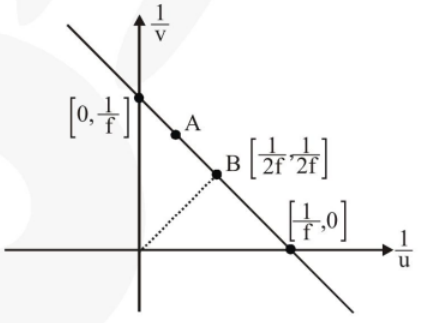
Q4. Figure shows two plane mirrors parallel to each other and an object O placed between them. Then the distance of the first three images from the mirror M_2 will be : (in cm)



(A) 5, 10, 15 (B) 5, 15, 30

(C) 5, 25, 35 (D) 5, 15, 25

Q5. Figure shows $\frac{1}{v}$ vs $\frac{1}{u}$ curve for convex mirror. Nature of image at point A is :-



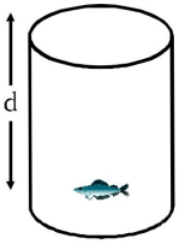
(A) Real, erect, magnified

(B) virtual, inverted, magnified

(C) virtual, inverted, diminished

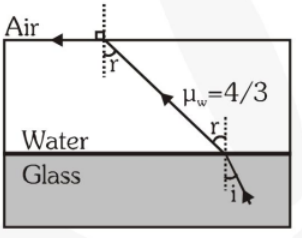
(D) virtual, erect, diminished

Q6. A fish floats in liquid with its eye at the centre of an opaque walled full tank of liquid of circular cross section. When the fish look upwards, it can see a fish-eye view of the surrounding scene i.e. it is able to view the entire space above the liquid surface. The diameter of the tank is 30 cm, and the critical angle for liquid is 37°. At what maximum depth below the surface of the liquid, must the fish be floating? $\left(\sin 37° = \frac{3}{5}\right)$



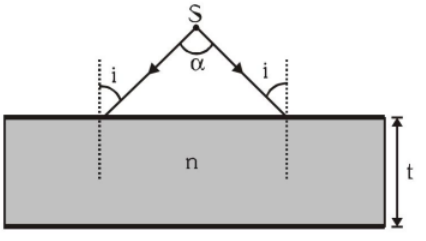
(A) 16 cm (B) 20 cm
(C) 11.25 cm (D) 25 cm

Q7. A ray of light is incident at the glass–water interface at an angle i, it emerges finally parallel to the surface of water, then the value of μ_g would be :-



(A) (4/3) sin i (B) 1/sin i
(C) 4/3 (D) 1

Q8. A diverging beam of light from a point source S having divergence angle α falls symmetrically on a glass slab as shown. The angles of incidence of the two extreme rays are equal. If the thickness of the glass slab is t and its refractive index is n, then the divergence angle of the emergent between them is :



(A) zero (B) α
(C) $\sin^{-1}$ (1/n) (D) 2 $\sin^{-1}$ (1/n)

Q9. A ray of light travelling in water is incident on its surface open to air. The angle of incidence is θ, which is less than the critical angle. Then there will be :-

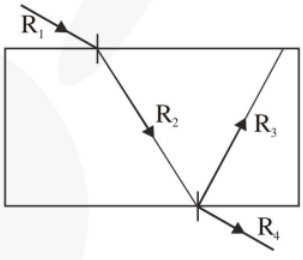
(A) Only a reflected ray and no refracted ray

(B) Only a refracted ray and no reflected ray

(C) A reflected ray and a refracted ray and the angle between them would be less than $180°-2\theta$.

(D) A reflected ray and a refracted ray and the angle between them would be greater than $180° - 2\theta$.

Q10. A ray R_1 is incident on the plane surface of the glass slab (kept in air) of refractive index $\sqrt{2}$ at an angle of incidence equal to the critical angle for this air glass system. The refracted ray R_2 undergoes partial reflection and refraction at the other surface. The angle between reflected ray R_3 and the refracted ray R_4 at that surface is :



(A) 45° (B) 135°
(C) 105° (D) 75°

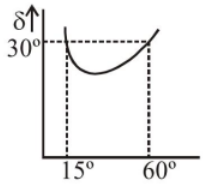
Q11. A ray of light passes through equilateral Prism (μ =1.5) such that angle of incidence is equal to angle of emergence and the later is equal to 3/4th of Prism angle. The angle of deviation is

(A) 60° (B) 30°
(C) 45° (D) 120°

Q12. Three prisms 1,2 and 3 have the prism angle A = 60°, but their refractive indices are respectively 1.4, 1.5 and 1.6.If δ_1, δ_2,δ_3, be their respective angles of deviation then :-

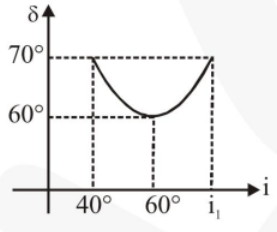
(A) $\delta_3 > \delta_2 > \delta_1$ (B) $\delta_1 > \delta_2 > \delta_3$
(C) $\delta_1 = \delta_2 = \delta_3$ (D) $\delta_2 > \delta_1 > \delta_3$

Q13. Figure shows graph of deviation δ versus angle of incidence for a light ray striking a prism. Angle of prism is :-



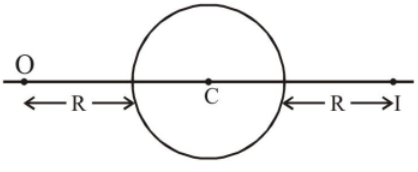
(A) 30°
(B) 45°
(C) 60°
(D) 75°

Q14. The curve of angle of incidence versus angle of deviation shown has been plotted for prism. The value of refractive index of the prism used is :



(A) $\sqrt{3}$
(B) $\sqrt{2}$
(C) $\frac{\sqrt{3}}{\sqrt{2}}$
(D) $\frac{2}{\sqrt{3}}$

Q15. In the given situation, a spherical transparent surface of radius R is placed. The refractive index of sphere is such that image for an object kept at distance R is formed at the same distance of the opposite side. What is refractive index of sphere ?



(A) 1.5
(B) 1.75
(C) 1.25
(D) 2

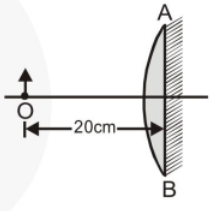
Q16. A concave lens of glass, refractive index 1.5 has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a :-

(A) convergent lens of focal length 3.5 R
(B) convergent lens of focal length 3.0 R
(C) divergent lens of focal length 3.5 R
(D) divergent lens of focal length 3.0 R

Q17. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now, this lens has been used to form the image of an object. At what distance from this lens, an object be placed in order to have a real image of the size of the object?

(A) 20 cm
(B) 30 cm
(C) 60 cm
(D) 80 cm

Q18. A point object is placed at a distance of 20 cm form a thin plano-convex lens of focal length 15 cm. If the plane surface is silvered, the image will from at :



(A) 60 cm from left of AB
(B) 30 cm from left of AB
(C) 12 cm from left of AB
(D) 60 cm from right of AB

Q19. If the refractive indices of crown glass for red, yellow and violet colours are 1.5140, 1.5170 and 1.5318 respectively and for flint glass these are 1.6434, 1.6499 and 1.6852 respectively, then the dispersive powers for crown and flint glass are respectively.

(A) 0.034 and 0.064
(B) 0.064 and 0.034
(C) 1.00 and 0.064
(D) 0.034 and 1.0

Q20. Lenses applied in achromatic combination have dispersive power in ratio of 5:3. If focal length of concave lens is 15cm, then focal length of other lens will be :

(A) -9cm (B) +9cm

(C) -12cm (D)+12 cm

Q21. A myopic person can not see objects lying beyond 2 m. The focal length and power of the lens required to remove this defect will be -

(A) 1 m & 0.5 D (B) – 2 m & – 0.5 D

(C) 0.5 m & 0.5 D (D) – 0.5 m & 0.5 D

Q22. A person's near point is 50 cm and his far point is 3m. Powers of the lenses he required for :

(i) for reading

(ii) for viewing distinct stars

(A) - 2D and 0.33 D (B) 2D and - 0.33 D

(C) - 2D and 3D (D) 2D and - 3 D

Q23. A telescope consisting of objective of focal length 60cm and a single lens eye piece of focal length 5cm is focussed at a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of 2° at the objective, then angular width of image will be.

(A) 10° (B) 24° (C) 50° (D) 1/6°

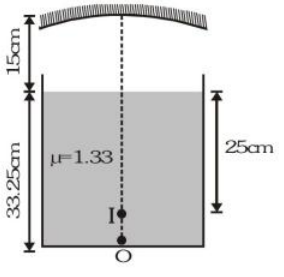
Q24. A container is filled with water (μ = 1.33) upto a height of 33.25 cm. A concave mirror is placed 15cm above the water level and the image of an object placed at the bottom is formed 25cm below the water level. The focal length of the mirror is :-

(A) 10 cm

(B) 15 cm

(C) 18.3 cm

(D) 25 cm



Q25. A plane mirror OA of length 2 m is kept along the line y = –x as shown in the figure.

An insect having velocity of $4\hat{i}$ cm/s is moving along the x-axis from far away. The time span for which the insect can see its image will be :-

(A) 50 sec (B) 25 sec

(C) $25\sqrt{2}$ sec (D) $50\sqrt{2}$ sec

Q26. There is a prism with refractive index equal to $\sqrt{2}$ and the refracting angle equal to 30°. One of the refracting surface of the prism is polished. A beam of monochromatic light will retrace its path if its angle of incidence over the first refracting surface of the prism is -

(A) 0° (B) 30°

(C) 45° (D) 60°

Q27. Figure A shows two identical plano-convex lenses in contact as shown. The combination has focal length 24 cm. Figure B shows the same with a liquid introduced between them. If refractive index of glass of the lenses is 1.50 and that of the liquid is 1.60, the focal length of the system in figure B will be :-

(A) –120 cm (B) 120 cm

(C) –24 cm (D) 24 cm

Q28. The size of the image of an object, which is at infinity, as formed by a convex lens of focal length 30cm is 2cm. If a concave lens of focal length 20cm is placed between the convex lens and the image at a distance of 26cm from the convex lens, calculate the new size of the image :–

(A) 1.25 cm (B) 2.5 cm

(C) 1.05 cm (D) 2 cm

Q29. A glass convex lens (μ_g = 1.5) has a focal length of 8 cm when placed in air. What would be the focal length of the lens when it is immersed in water (μ_w = 1.33)

(A) 2 m (B) 4 cm

(C) 16 cm (D) 32 cm

Q30. A thin Prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin Prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of Prism P_2 is:-

(A) 5.33° (B) 4°

(C) 3° (D) 2.6°

Q31. Angle of prism is A and its one surface is silvered. Light rays falling at an angle of incidence 2A on first surface return back through the same path after suffering reflection at second silvered surface. Refractive index of the material of prism is:-

(A) 2 sin A (B) 2 cos A

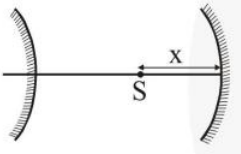
(C) $\frac{1}{2}$ cosA (D) tanA

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A convex mirror and a concave mirror are placed on the same optic axis, separated by distance $L = 20$ cm. The radius of curvature of each mirror has a magnitude of 12 cm. A light source is located at a distance x from the concave mirror as shown in figure.



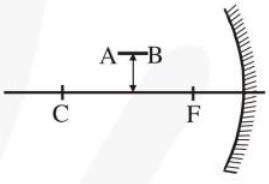
(A) The rays from the source return to the source after reflecting first from the convex mirror and then from the concave mirror for $x = 8$ cm.

(B) The rays from the source return to the source after reflecting first from the convex mirror and then from the concave mirror for $x = 16$ cm.

(C) If the rays from the source return to the source after first reflecting from the concave mirror and then from the convex mirror for $x = 8$ cm.

(D) If the rays from the source return to the source after first reflecting from the concave mirror and then from the convex mirror for $x = 16$ cm.

Q2. An object AB is placed parallel and close to the optical axis between focus F and centre of curvature C of a converging mirror of focal length f as shown in figure.



(A) Image of A will be closer than that of B from the mirror.

(B) Image of AB will be parallel to the optical axis.

(C) Length of image is equal to AB.

(D) Length of image is more than AB.

Q3. When a ray of light of frequency 6×10^{14} Hz travels from water of refractive index 4/3 to the glass of refractive index 8/5, its :-

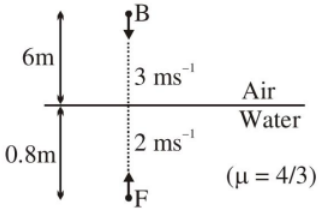
(A) frequency decreases to 5/6 of its initial value

(B) speed decreases to 5/6 of its initial value

(C) wavelength decreases to 5/6 of its initial value

(D) speed increases to 6/5 of its initial value

Q4. A fish, F in the pond, is at a depth of 0.8 m from water surface and is moving vertically upwards with velocity 2 ms^{-1}. At the same instant, a bird B is at a height of 6 m from water surface and is moving downwards with velocity 3 ms^{-1}. At this instant both are on the same vertical lines as shown in the figure. Which of the following statement(s) is(are) correct?



(A) Height of B, observed by F (from itself) is equal to 8.00 m.

(B) Depth of F, observed by B (from itself) is equal to 6.60 m.

(C) Velocity of B, observed by F (relative to itself) is equal to 5.00 ms^{-1}.

(D) Velocity of F, observed by B (relative to itself) if equal to 4.50 ms^{-1}.

Q5. In the figure (a) light is incident at an angle θ which is slightly greater than the critical angle. Now, keeping the incident fixed a slab of refractive index n_3 is placed on surface AB in fig. (b). Which of the following statements are correct :

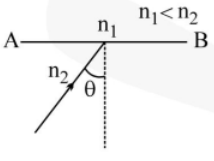


Fig. (a)

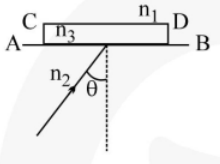


Fig. (b)

(A) Total internal reflection occurs at AB for $n_3 < n_1$

(B) Total internal reflection occurs at AB for $n_3 > n_1$

(C) The ray will return back to the same medium for all values of n_3

(D) Total internal reflection occurs at CD for $n_3 < n_1$

Q6. For the refraction of light through a prism

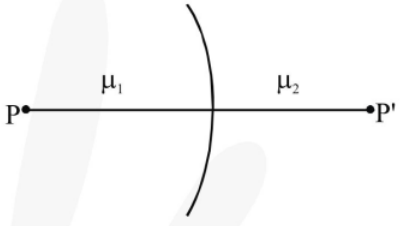
(A) For every angle of deviation there are two angles of incidence.

(B) The light travelling inside an equilateral prism is necessarily parallel to the base when prism is set for minimum deviation.

(C) There are two angles of incidence for maximum deviation.

(D) Angle of minimum deviation will increase if refractive index of prism is increased keeping the outside medium unchanged if $\mu_{Prism} > \mu_{Surrounding}$.

Q7. Two refracting media are separated by a spherical interface as shown in the figure. *PP'* is the principal axis, μ_1 and μ_2 are the refractive indices of medium of incidence and medium of refraction respectively, then:-



(A) if $\mu_2 > \mu_1$, then there cannot be a real image of real object.

(B) if $\mu_2 > \mu_1$, then there cannot be a real image of virtual object.

(C) if $\mu_1 > \mu_2$, then there cannot be a virtual image of virtual object.

(D) if $\mu_1 > \mu_2$, then there cannot be a real image of real object.

Q8. Optical axis of a thin equi-convex lens is the X-axis. The co-ordinate of a point object and its image are (–20 cm, 1 cm) and (25 cm, –2 cm) respectively :

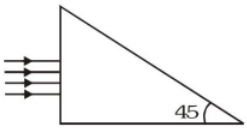
(A) The lens is located at x = 5 cm

(B) The lens is located at x = –5 cm

(C) The focal length of the lens is 10 cm

(D) The focal length of the lens is 15 cm

Q9. A beam of light consisting of red, green and blue and is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will



(A) Separate part of the red color from the green and blue colors

(B) Separate part of the blue color from the red and green colors

(C) Separate all the three colors from the other two colors

(D) Not separate even partially, any colors from the other two colors

Q10. Chromatic aberration will be absent if for two thin lenses in contact :-

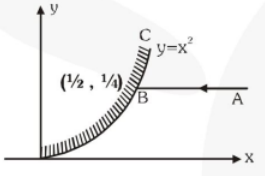
(A) $(\omega_1/f_1) + (\omega_2/f_2) = 0$

(B) $(\omega_1/f_2) + (\omega_2/f_1) = 0$

(C) $(f_1/\omega_2) + (f_2/\omega_1) = 0$

(D) $(\omega_1/\omega_2) + (f_1 + f_2) = 0$

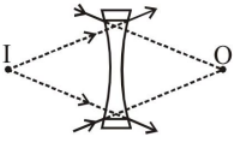
Q11. In the given figure, incident ray AB is parallel to x-axis. The angle of reflection is:-

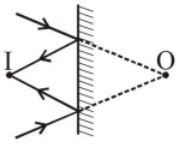


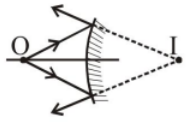
(A) 30° (B) 60°

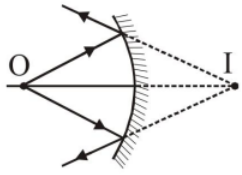
(C) 45° (D) None of these

Q12. The nature of object and image given with each of the optical condition is shown. Choose the **correct** option(s) :-

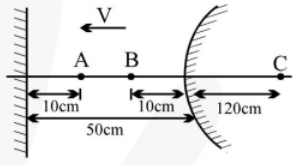
(A) Virtual image  Virtual object

(B) Virtual image  Real object

(C) Real object  Virtual image

(D) Real object  Virtual image

Q13. In the figure shown consider the first reflection at the plane mirror and second at the convex mirror. AB is object.



(A) The second image is real, inverted of $1/5^{th}$ magnification w.r.t AB

(B) The second image is virtual and erect with magnification1/5 w.r.t AB

(C) The second image moves towards the convex mirror

(D) The second image moves away from the convex mirror.

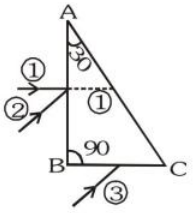
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 14 to 16

ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the value that will be required so that ray (1) grazes the face AC.

Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) to have the same frequency, answer the following questions.



Q14. Refractive index of the material of prism is:-

(A) $\sqrt{2}$ (B) 2

(C) 1.5 (D) 2.2

Q15. Angle of incidence of ray (2) is :-

(A) 0° (B) 45°

(C) 60° (D) 90°

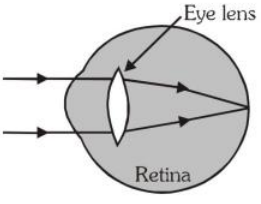
Q16. Deviation suffered by ray (3) is :-

(A) 60° (B) 30°

(C) 90° (D) 120°

Paragraph for Question No. 17 to 21

The ciliary muscles of eye control the curvature of the lens in the eye and hence can alter the effective focal length of the system. When the muscles are fully relaxed, the focal length is maximum. When the muscles are strained the curvature of lens increases (that means radius of curvature decreases) and focal length decreases. For a clear vision the image must be on retina. The image distance is therefore fixed for a clear vision and it equals the distance of retina from eye–lens. It is about 2.5 cm for a grown–up person.



A person can theoretically have clear vision of objects situated at any large distance from the eye. The smallest distance at which a person can clearly see is related to minimum possible focal length. The ciliary muscles are most strained in this position. For an average grown up person minimum distance of object should be around 25 cm.

A person suffering from eye defects uses spectacles (Eye glass). The function of lens of spectacles is to form the image of the objects within the range in which person can see clearly. The image of the spectacle–lens becomes object for eye–lens and whose image is formed on retina.

The number of spectacle–lens used for the remedy of eye defect is decided by the power of the lens required and the number of spectacle–lens is equal to the numerical value of the power of lens with sign. For example power of lens required is +3D (converging lens of focal length $\frac{100}{3}$ cm) then number of lens will be +3).

For all the calculations required you can use the lens formula and lens maker's formula. Assume that the eye lens is equiconvex lens. Neglect the distance between eye lens and the spectacle lens.

Q17. Minimum focal length of eye lens of a normal person is :

(A) 25 cm (B) 2.5 cm

(C) $\frac{25}{9}$ cm (D) $\frac{25}{11}$ cm

Q18. Maximum focal length of eye lens of normal person is:

(A) 25 cm (B) 2.5 cm

(C) $\frac{25}{9}$ cm (D) $\frac{25}{11}$ cm

Q19. A near–sighted man can clearly see object only upto a distance of 100 cm and not beyond this. The number of the spectacles lens necessary for the remedy of this defect will be:

(A) + 1 (B) –1

(C) + 3 (D) –3

Q20. A farsighted man cannot see object clearly unless they are at least 100 cm from his eyes. The number of the spectacles lens that will make his range of clear vision equal to an average grown up person

(A) + 1 (B) –1

(C) + 3 (D) –3

Q21. A person who can see object clearly from distance 10 cm to ∞, then we can say that the person is:

(A) Normal sighted person

(B) Near–sighted person

(C) Far–sighted person

(D) A person with exceptional eye having no eye defect

Paragraph for Question No. 22 to 24

Q22. If an astronomical telescope has objective and eye-pieces of focal lengths 200 cm and 4 cm respectively, then the magnifying power of the telescope for the normal vision is :-

(A) 42 (B) 50 (C) 58 (D) 204

Q23. In the above question the length of the telescope for normal vision, is :-

(A) 204 cm (B) 200 cm

(C) 196 cm (D) 203.45 cm

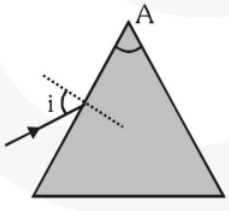
Q24. In the above question magnifying power of the telescope for distinct vision is :-

(A) 42 (B) 50 (C) 58 (D) 204

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q25. For a prism of refracting angle A and refractive index 2. Assume rays are incident at all angles of incidence $0° \le i \le 90°$. Ignore partial reflection.



Column–I	**Column–II**
(A) A = 15°	(P) All rays are reflected back from the second surface.
(B) A = 45°	(Q) All rays are refracted into air from the second surface
(C) A = 70°	(R) Some rays are reflected back from second surface
(D) A = 50°	(S) Some rays are refracted into air from the second surface

NUMERICAL TYPE

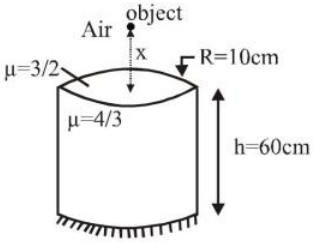
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q26. A concave mirror of focal length 10 cm and convex mirror of focal length 15 cm are placed facing each other 40 cm apart. A point object is placed between the mirror on their commen axis and 15 cm from the concave mirror. Find the position of image produced by the reflection first at concave mirror and then at convex mirror.

Q27. A pin is placed 10 cm in front of a convex lens of focal length 20 cm, made of material having refractive index 1.5. The surface of the lens farther away from the pin is silvered and has a radius of curvature 22 cm. Determine the position of the final image. Is the image real or virtual ?

Q28. A glass convex lens of refractive index $\frac{3}{2}$ has got a focal length equal to 0.3 m. Find the focal length of the lens if it is immersed in water of refractive index $\frac{4}{3}$.

Q29. In the arrangement shown given that a biconvex lens of radius of curvature equal to 10 cm and concave mirror has focal length equal to 20cm. Then find the distance 'x' such that the final image formed by the system coincides with the object.



SUBJECTIVE TYPE

Q30. A object is placed in front of a convex mirror at a distance of 50 cm. A plane mirror is introduced covering the lower half of the convex mirror. If the distance between the object and the plane mirror is 30 cm, it is found that there is no parallax between the images formed by the two mirrors. What is the radius of curvature of the convex mirror? Also calculate magnification produced by convex mirror.

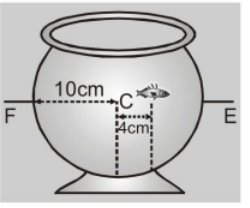
Q31. A ray of light from a denser medium strikes a rarer medium at an angle of incidence i. If the reflected and refracted rays are mutually perpendicular to each other, what is the value of critical angle ?

Q32. A beam of light converges to a point P. A lens is placed in the path of the convergent beam 12 cm from P. At what point does the beam converge if the lens is :-
(a) a convex lens of focal length 20 cm.
(b) a concave lens of focal length 16 cm.

Q33. A 20 cm thick glass slab of refractive index 1.5 is kept infront of a plane mirror. An object is kept in air at a distance 40 cm from the mirror. Find the position of image relative to mirror. What is effect of separation between glass slab and the mirror on image.

Q34. A man with normal near point 25 cm reads a book with small print using a magnifying glass, a thin convex lens of focal length 5 cm.
(a) What is the closest and fartest distance at which he can read the book when viewing through the magnifying glass?
(b) What is the maximum and minimum magnifying power possible using the above simple microscope ?

Q35. In a thin spherical fish bowl of radius 10 cm filled with water of refractive index $\left(\frac{4}{3}\right)$, there is a small fish at a distance 4 cm from the centre C as shown in Figure. Where will the fish appear to be, if seen from (a) E and (b) F (neglect the thickness of glass) ?



Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	B	A	A	B	A	B	B	C	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	B	A	C	A	C	C	B	B	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	A	A	D	A	C	B	A	A	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	D	D	B	D	A	D	B	C	A
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
B	D	D	A	B	B	C	C	B	C
51									
B									

Ex-1A

1. A,B **2.** B,C **3.** A,D **4.** A,D **5.** A,B,C

6. A,B,C,D **7.** A,D **8.** A,B,D **9.** B **10.** B

11. C

12. (A) → (Q), (B) → (P), (C) → (S), (D) → (R)

13. (A)–Q; (B)–P; (C)–S; (D)–R

14. (A)→(P,S,T);(B)→(Q,S);(C)→(Q, R) (D)→(Q, R)

15. (A)–R; (B)–Q; (C)–P; (D)–S

16. (A)–Q, S (B)–P,Q,R,S (C)–P,Q,S (D)–S

17. 1.33 **18.** 0.02. **19.** 2.12×10^8 ms^{-1}

20. (a) v = –20 cm, $m = -\left[-\frac{20}{10}\right] = +2$ (b) v = +60 cm, $m = -\left[+\frac{60}{30}\right] = -2$

21. (a) 7.5 cm (b) $m = \left[\frac{h_2}{h_1}\right] = \frac{v}{u} = \frac{-30}{-7.5} = 4$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	D	A	C	C	B	B	B	C	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	A	B	A	D	A	A	C	A	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	B	C	D	C	A	B	D	C
31									
B									

Ex-2A

1. A,C **2.** A,D **3.** B,C **4.** B,D **5.** A,C

6. B,C,D **7.** A,C **8.** B,C **9.** A **10.** A

11. C **12.** A,C,D **13.** B,C **14.** B **15.** D

16. D **17.** D **18.** B **19.** B **20.** C

21. D **22.** B **23.** A **24.** C

25. (A)–Q, (B)–R,S (C)–P (D)–R, S

26. 6 cm **27.** 11 cm **28.** 1.2 m **29.** 22.5 cm

30. $R = 2f = 25$ cm, $m = +\left[\frac{1}{5}\right]$ **31.** $\theta_C = \sin^{-1}\left[\frac{1}{\cot i}\right] = \sin^{-1}(\tan i)$

32. (a) $v = \frac{60}{8} = 7.5$ cm (b) $v = 48$ cm

33. 11 cm, Real

34. (a) $(MP)_{min} = \frac{-25}{-5} = 5$ (b) $(MP)_{max} = \frac{-25}{-(25/6)} = 6$

35. (a) 5.3 cm (b) 16.154 cm